MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Short Answer Type-I question of 2 marks.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (5 × 1 = 5 Marks)

Q1. If $5\tan\beta = 4$, then $\frac{5\sin\beta - 2\cos\beta}{5\sin\beta + 2\cos\beta}$ is equal to: [1]

(A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) 6

Q2. The value of $\frac{2\tan 30^\circ}{1+\tan^2 30^\circ}$ is equal to: [1]

(A) $\sin 60^\circ$ (B) $\cos 60^\circ$
(C) $\tan 60^\circ$ (D) $\sin 30^\circ$

Q3. The value of $\frac{1-\tan^2 45^\circ}{1+\tan^2 45^\circ}$ is equal to: [1]

(A) $\tan 90^\circ$ (B) 1 (C) $\sin 45^\circ$ (D) 0

Q4. If $\sin 2A = 2\sin A$ is true, then the value of A is: [1]

(A) 0° (B) 30° (C) 45° (D) 60°

Q5. **Assertion (A):** The value of $\cos\theta$ increases as θ increases from 0° to 90°.
Reason (R): $\cos 0^\circ = 1$ and $\cos 90^\circ = 0$, so $\cos\theta$ decreases as θ increases. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type-I (1 × 2 = 2 Marks)

Q6. If $\sin(A-B) = \frac{1}{2}$ and $\cos(A+B) = \frac{1}{2}$, where $0^\circ < A+B \leq 90^\circ$ and $A > B$, find the values of acute angles A and B. [2]

SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q7. Prove that: $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\csc\theta$. [3]

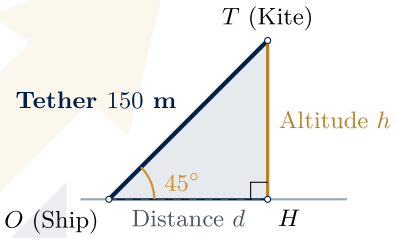
Q8. Prove that: $(\csc A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$. [3]

Q9. Prove that: $\left(\frac{1+\tan^2 A}{1+\cot^2 A}\right) = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$. [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Maritime Kite Propulsion (Skysails Engineering)

A modern cargo ship uses an auxiliary towing kite system ('Skysails') to reduce fuel consumption. The towing kite flies high above the ocean surface, attached to the bow by a high-strength tether cable OT of length 150 m. The cable makes an angle of $\theta = 45°$ with the horizontal deck of the ship.



Based on the above geometric model, answer the following questions:

(i) Find the vertical operating altitude h of the towing kite above the deck. [1]

(ii) Find the horizontal forward pulling distance d from the attachment point to the kite's vertical plumb line. [1]

(iii) If the total tension force along the tether cable is $T = 80$ kN, calculate the horizontal forward propulsive force component $F_x = T\cos 45°$ and vertical lift force $F_y = T\sin 45°$. [2]

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